

Jiahui Zhang

Email: zjh68688@gmail.com Mob: +1 2066875212
1714 Gentry Square Ln, APT 203, Champaign, IL 61821

EDUCATION

University of Washington

Master of Science in Statistics

GPA: 3.86/4.0

Seattle, USA

Sep 2023 – Mar 2025

University of Liverpool

Bachelor of Science with Honours in Mathematics (Class I)

GPA: 83/100; WES evaluated GPA: 3.94/4.0; Rank: top 5%

Liverpool, England

Sep 2021 – Jun 2023

Xi'an Jiaotong-Liverpool University

Bachelor of Science with Honours in Applied Mathematics (Class I)

GPA: 78/100; Rank: top 8%

Suzhou, China

Sep 2019 – Jun 2021

PUBLICATIONS

Andrews A, Harrar DB, Zelleke T, **Zhang J**, Zhang B, Barber J, et al.

Electroencephalogram features distinguish cases of cerebral malaria among Malawian children with fever and coma. *Clinical Infectious Diseases*. 2025 Jul 30; ciaf316. doi:10.1093/cid/ciaf316. Epub ahead of print.

Takle M, Andrews A, Riggle BA, Zelleke T, Harrar D, **Zhang J**, Zhang B, et al.

Using electroencephalography to assess coma etiology in children with retinopathy-negative cerebral malaria. *American Journal of Tropical Medicine and Hygiene*. 2025 Jul 22; tpmd250377. doi:10.4269/ajtmh.25-0377. Epub ahead of print.

Jiahui Zhang, Ziyu Yu, Xingcai Zhang, Haohan Wang.

Stratifying Alzheimer's Disease by Patient-Specific Genetic Signatures Reveals Cognition-Linked and Cross-Disease Heterogeneity.

(Manuscript submitted)

Jiahui Zhang, Chan Park, Ruoqing Zhu.

Covariate similarity driven multi-graph approach for treatment effect estimation under interference.

(Manuscript under preparation)

RESEARCH EXPERIENCE

Covariate similarity driven multi-graph approach for treatment effect estimation under interference |

Research Assistant

May 2025 - Present

Advisor: Ruoqing Zhu (Professor, Department of Statistics, University of Illinois Urbana-Champaign)

- Developed a covariate similarity driven multi-graph approach that models interference through covariate-based similarity graphs, extending applicability to both known and unknown-network settings.
- Formulated multi-dimensional exposure vectors to capture heterogeneous spillovers, negative peer effects, and nonlinear interference via linear and neural network forms (incorporating tanh + PReLU activation functions) for broader application.
- Introduced ATEs based on treatment intensity to evaluate treatment effects under varying coverage levels, generalizing beyond traditional all-or-none contrasts.
- Conducted large-scale simulations (ER and covariate-driven graphs, $n = 4000$) demonstrating that new approach achieves near-zero bias and outperforms SNIPE, LS, and difference-in-means estimators.
- Currently in the process of finalizing a research manuscript for journal submission.

Heteroscedastic Personalized Regression Framework for Individualized Genomic Analysis |

Research Assistant

Aug 2025 - Present

Advisor: Haohan Wang (Assistant Professor, School of Information Sciences, University of Illinois Urbana-Champaign)

- Developed the Heteroscedastic Personalized Regression (Het-PR) framework, reformulating personalized modeling as a linear mixed-effects model with sample-specific variance terms to estimate individualized SNP effect profiles.

- Conducted extensive simulations demonstrating that Het-PR outperforms population-level models: it reduced estimation MSE by ~30–60% and achieved 82–94% AUROC in identifying subject-specific genetic effects, while enabling recovery of latent patient subgroups (~90–100% accuracy via hierarchical clustering).
- Applied Het-PR to the ADNI cohort (GWAS data), stratifying Alzheimer’s patients into two genetically distinct clusters with consistently divergent cognitive performance across five domains, thereby linking fine-scale genetic signatures to the severity of cognitive impairment.
- Identified cluster-specific SNP sets and performed pathway enrichment analysis, revealing distinct functional architectures.
- Discovered that variants previously associated with epilepsy powerfully separate the two AD clusters, indicating a shared cross-disease genetic axis and suggesting new genetic links between Alzheimer’s and epilepsy.

Modeling the Per-exposure Efficacy of Immune-based Therapies for HIV-1 Prevention | Research Assistant May 2024 - Present

Advisor: Bo Zhang (Assistant Professor, Fred Hutch; Affiliate Assistant Professor, Department of Biostatistics, University of Washington)

- Developed a microsimulation model to predict HIV-1 acquisition risks following neutralizing antibody (bnAb) infusion, incorporating viral exposures, individual pharmacokinetics, and viral sensitivity.
- Calibrated the model with Bayesian rejection sampling using AMP trial data, estimating key parameters governing infection risk and neutralization efficacy.
- Enhanced inference by implementing ABC-MCMC, integrating both case counts and IC80 (virus neutralization sensitivity) distributions to improve posterior accuracy and acceptance rates.
- Designed a Shiny app (<https://zjh68688.shinyapps.io/BayesShiny/>) to simulate trial scenarios, producing efficacy and sensitivity distribution plots.

Classifying Cerebral Malaria from Non-cerebral Malaria Using EEG Data Based on Machine Learning Methods | Research Assistant Oct 2023 - Apr 2025

Advisor: Bo Zhang (Assistant Professor, Fred Hutch; Affiliate Assistant Professor, Department of Biostatistics, University of Washington)

- Conducted exploratory data analysis (EDA) on EEG signals, including biomarker selection, missing data imputation, and categorical encoding.
- Trained ensemble machine learning models (GLM, GAM, Random Forest) with ten-fold cross-validation across baseline, qualitative, and quantitative EEG features.
- Evaluated predictive performance using ROC curves and AUC, showing strong discrimination between cerebral malaria (CM) and non-malarial cases, including retinopathy-negative CM.
- Co-authored two peer-reviewed manuscripts in Clinical Infectious Diseases and American Journal of Tropical Medicine and Hygiene.

Evaluating Causal Relation between Job Training Programs and Income with Multiple Causal Methods | Project Team Leader Apr 2024 - Jun 2024

Advisor: Thomas Richardson (Professor, Department of Statistics, University of Washington)

- Implemented inverse probability weighting, outcome regression, doubly robust, and DID estimators on experimental (NSW) and observational (PSID) datasets to estimate treatment effects.
- Compared estimates against the experimental benchmark, identifying overlap violations in the PSID data that led to unstable or negative ATEs, while DiD produced positive but inflated effects.

LEADERSHIP AND ACTIVITIES

XJTLU Mathematics Society | Vice President 2019-2020

- Organized academic lectures by inviting professors and senior students as guest speakers to enrich the academic community; published materials on the society’s social media account.

SKILLS AND INTERESTS

Programming: Python, R, C++, Latex, Java, MATLAB

Language: Mandarin Chinese (native), English (fluent)

Interests: badminton, running, movies, travel